CHAPTER 1 • GRADE 4 • WEEK 1

Energy Causes Change

DISCOVERY • EVIDENCE

Where can we observe evidence of energy causing change?

ENERGY → CHANGE → ?



Big Question

BIG QUESTION

Where can we observe evidence of energy causing change?

Keep it in mind all chapter. We will come back to it.

Think: What is ONE change you have seen happen?

ENERGY → CHANGE → ?

Big Question — My first thinking

Guided Notes p. 1

Where can we observe evidence of energy causing change?

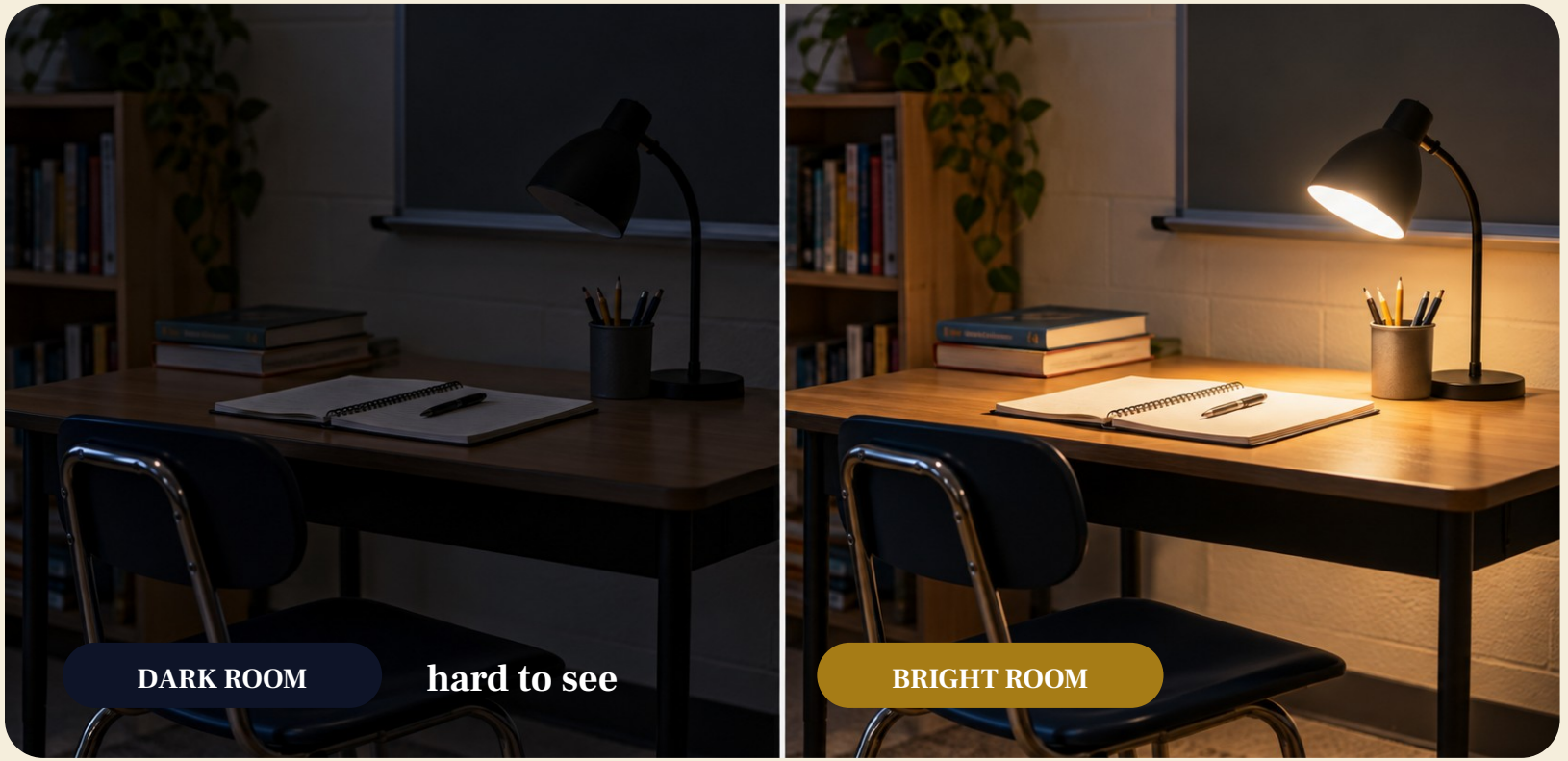
My first thinking

Write 1–2 quick ideas. No wrong answer yet.

OBSERVE

What changed?

Reader p. 1 — you walk into a dark room and flip a switch.



Something became different.

That is a change.

light fills the room

TURN & TALK

I notice… I wonder…



I NOTICE

The room is different now.
Light is everywhere.



I WONDER

What made the light change so fast?

Share one notice and one wonder with a partner.

OBSERVE

Electric Pencil Sharpener



BEFORE

Observe the pencil BEFORE we sharpen it.

A dull pencil — tip is flat, and it is quiet.

Look closely. What do you see? What do you hear?

NEXT: SHARPENER DEMO

PREDICT

Predict: what will change?

Think before we turn it on.

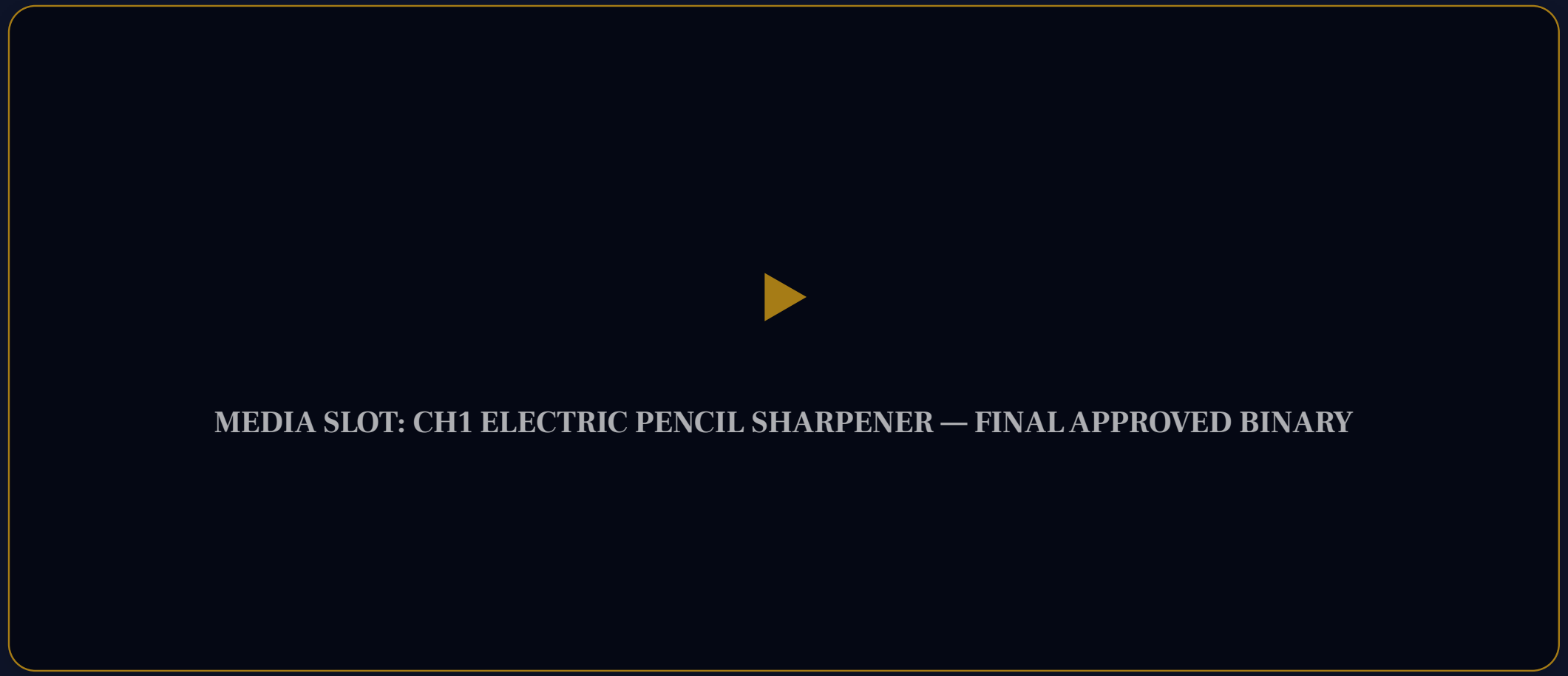
When the sharpener runs, the pencil will…

get _____

Make a prediction. Say it to your partner.

WATCH CLOSELY

Electric Pencil Sharpener Demo



What changes do you see or hear?

If the video is unavailable, use the live demo or continue to the evidence slide.

OBSERVE

What changed?





CHANGES

- the tip is sharper
- the pencil is shorter
- the sharpener makes a sound

The changes are clues that energy was involved.

Vocabulary

KEY WORD

energy



the ability to cause change

Reader p. 1

RECITE

Vocabulary

KEY WORD

change



to become different

Reader p. 1

Vocabulary — Energy & Change

Guided Notes • Reader p. 1

energy — the ability to cause _____ .

change — to become _____ .

Write the missing word in each blank.

Vocabulary — Energy & Change

Guided Notes • Reader p. 1

energy — the ability to cause change .

change — to become different .

change • different

OBSERVE

Look at the seashore

Reader p. 2 — the seashore changes all around us.

Waves move.
Sand shifts.
Water makes sound.

ENERGY → CHANGE → ?

What are these changes a clue about?



N1 — Energy Causes Change

Guided Notes • Reader pp. 1–2



Energy has the ability to cause change.

When you see a change, you are seeing _____ of energy.

Write the missing word. Leave the last box of your model empty for now.

N1 — Energy Causes Change

Guided Notes • Reader pp. 1–2



Energy has the ability to cause change.

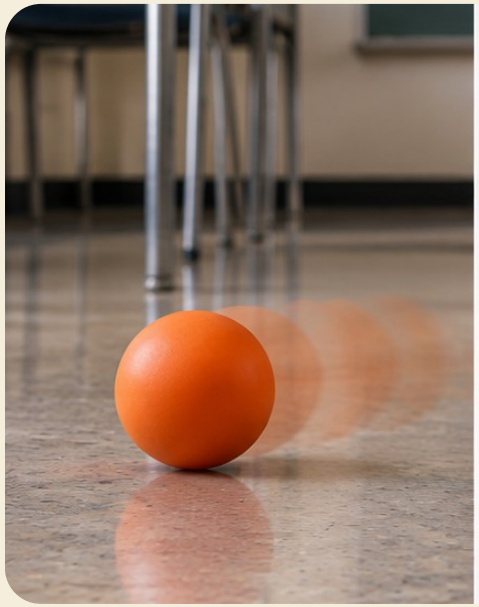
When you see a change, you are seeing evidence of energy.

Now write the same word in the last box of your model.

OPTIONAL MEDIA: ENERGY DETECTIVE

Energy has different forms

Where can we observe evidence of energy causing change?











MOTION

SOUND

LIGHT

HEAT

ELECTRICAL ENERGY

Say the five forms together.

OBSERVE

Five forms of energy

MOTION

SOUND

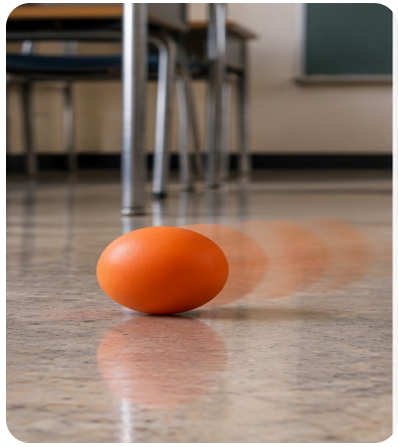
LIGHT

HEAT

ELECTRICAL ENERGY

OBSERVE

Energy has different forms



motion

A rolling ball moves.



sound

A bell makes sound.



light

A lamp makes an area brighter.



heat

A warm mug gives off heat.



electrical energy

A fan uses electrical energy to operate.

N2 — Energy Occurs in Different Forms

Guided Notes • Reader p. 3

Write one example that shows a change for each form.

FORM	ONE EXAMPLE THAT SHOWS A CHANGE
motion	
sound	
light	
heat	
electrical energy	

N2 — Energy Occurs in Different Forms

Guided Notes • Reader p. 3

SAMPLE ANSWERS — YOUR EXAMPLES MAY DIFFER

FORM	ONE EXAMPLE THAT SHOWS A CHANGE
motion	a ball moves across the floor
sound	a bell rings
light	a dark area becomes brighter
heat	something becomes warmer
electrical energy	a fan begins moving

RECITE

Vocabulary

KEY WORD

stored energy



energy that has the ability to cause change at a later time

Reader p. 4

Vocabulary — Stored Energy

Guided Notes • Reader p. 4

stored energy — energy that has the ability to cause change at a _____ time.

Write the missing word in the blank.

Vocabulary — Stored Energy

Guided Notes • Reader p. 4

stored energy — energy that has the ability to cause change at a later time.

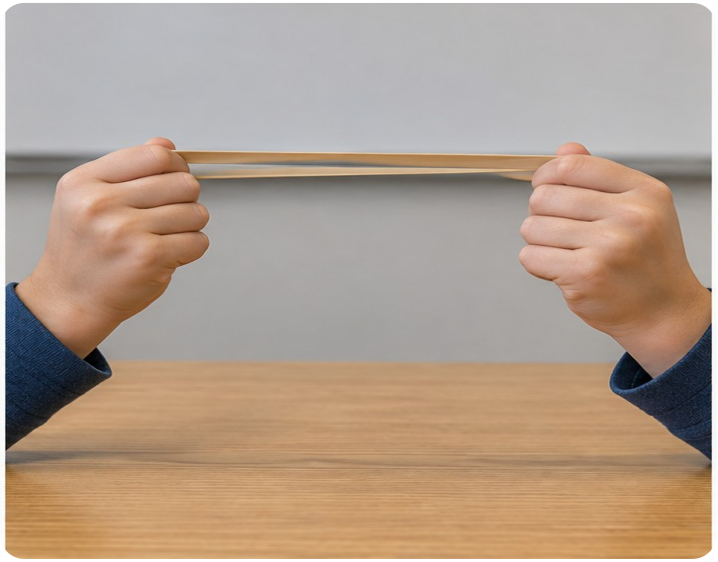
later

OBSERVE

Objects can store energy



battery (in a flashlight)



stretched rubber band



raised object (apple)

What change could happen later?

N3 — Objects Can Store Energy

Guided Notes • Reader pp. 4–5

STORED ENERGY → **RELEASE** → **CHANGE**

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	
stretched rubber band	
raised object (apple)	

Objects also have stored energy because of their _____ .

N3 — Objects Can Store Energy

Guided Notes • Reader pp. 4–5

STORED ENERGY → **RELEASE** → **CHANGE**

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	the flashlight lights up
stretched rubber band	it snaps and makes a sound
raised object (apple)	it falls down

Objects also have stored energy because of their position .

position • Sample answers shown — yours may differ.

Recognizing Evidence

What changed? What is the evidence?



LIGHT



PENCIL



MOTION

Describe only what you can observe.

N4 — Recognizing Evidence

Guided Notes • Reader pp. 1–5

Draw a change you observed. Show what it looked like BEFORE and AFTER.

BEFORE



AFTER

What changed?

Check Your Thinking

Guided Notes • Reader pp. 1–5

How do we know energy is involved if we cannot see energy itself?

Start with: “We know because we see …”

Check Your Thinking

Guided Notes • Reader pp. 1–5

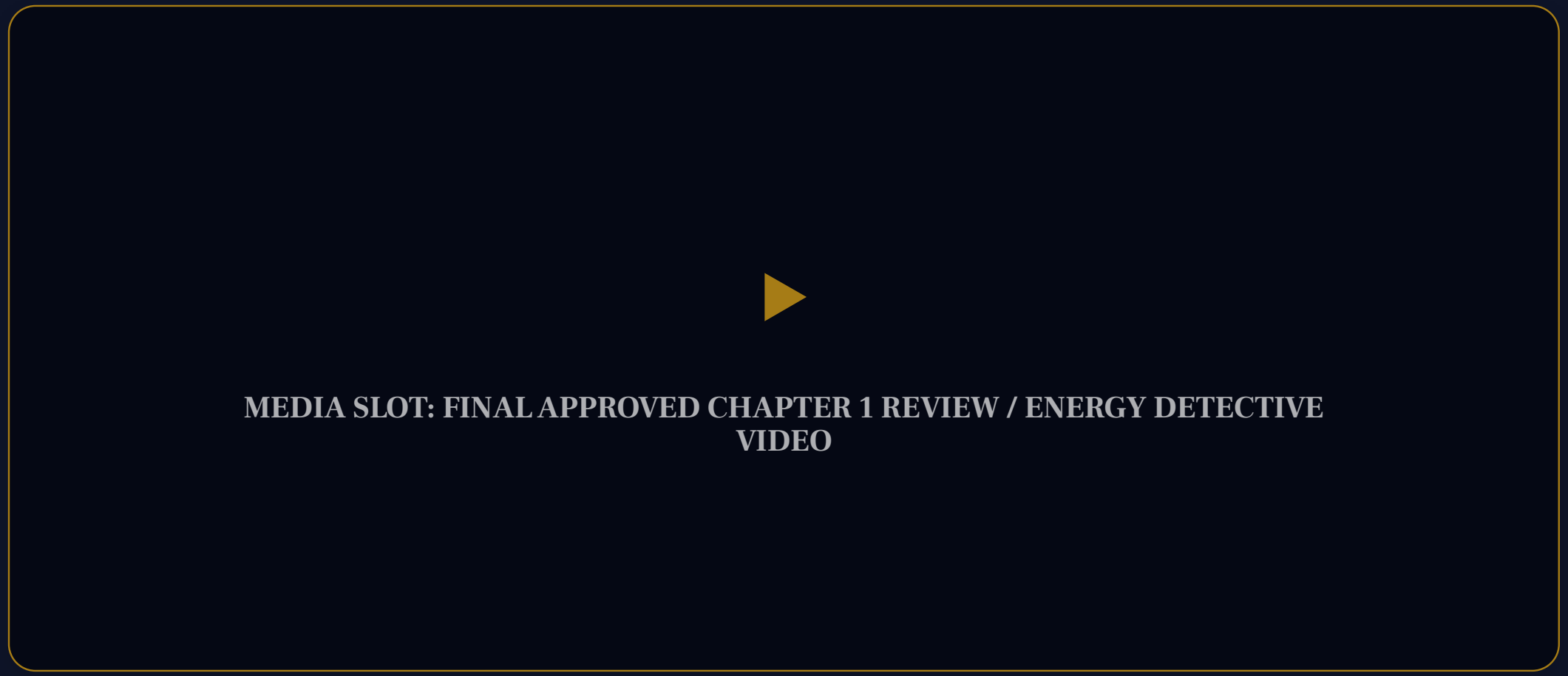
SAMPLE

“We know because we see a change, like a ball starting to move or a light turning on.”

A change you observe is evidence of energy.

OPTIONAL • SKIP IF SHORT ON TIME

CHAPTER 1 OPTIONAL REVIEW



Listen for energy, change, and evidence.

The Big Question Revisited

Where can we observe evidence of energy causing change?

MY FIRST THINKING

Look back at page 1.

NOW I THINK

ENERGY → CHANGE → EVIDENCE

VOTE

Review 1 — concept

Energy is best described as —

A a visible glow

B something only batteries have

C the ability to cause change

D a kind of fuel

CHECK

Review 1 — answer

Energy is best described as —

A a visible glow

B something only batteries have

✓ C **the ability to cause change**

D a kind of fuel

Energy is the ability to cause change (Reader p. 1).

VOTE

Review 2 — visual

A pencil is dull, then you sharpen it. What change do you observe?



A. It gets shorter and sharper

B. It gets longer

C. It changes color

D. It disappears

CHECK

Review 2 — answer

A pencil is dull, then you sharpen it. What change do you observe?



✓ A **It gets shorter and sharper**

B It gets longer

C It changes color

D It disappears

The pencil becomes shorter and sharper — a change.

VOTE

Review 3 — scenario

Which object has stored energy that can cause a change LATER?



A. A ball already rolling

B. A toy already moving

C. A flashlight already on

D. A stretched rubber band

Review 3 — answer

Which object has stored energy that can cause a change LATER?



A — A ball already rolling

B — A toy already moving

C — A flashlight already on

✓ D — **A stretched rubber band**

The rubber band stores energy; it changes when released.

VOTE

Review 4 — comparison

A carousel spins and plays music. Which two forms of energy come from the spinning and the music?

A only light

B motion and sound

C motion and silence

D sound and stillness

CHECK

Review 4 — answer

A carousel spins and plays music. Which two forms of energy come from the spinning and the music?

A only light

✓ B **motion and sound**

C motion and silence

D sound and stillness

The spinning is motion; the music is sound (Reader p. 3).

VOTE

Review 5 — concept

How do we know energy is involved if we cannot see it?

A. We observe a change happen

B. Energy can always be seen

C. We guess without evidence

D. Nothing can change

CHECK

Review 5 — answer

How do we know energy is involved if we cannot see it?

✓ A **We observe a change happen**

B Energy can always be seen

C We guess without evidence

D Nothing can change

We see a change — like movement — as evidence.

VOTE

Review 6 — scenario

A battery sits still in a flashlight. Does it have energy?



- **A** No — still objects have no energy
- **B** Only when it is moving
- **C** Yes — stored energy that can cause change later
- **D** Only when the flashlight is on

CHECK

A battery sits still in a flashlight. Does it have energy?

✓ C

Yes — stored energy that can cause change later

A battery stores energy. It can cause a change later.

ENERGY → CHANGE → EVIDENCE

Look for the change. Use the change as evidence that energy was involved.

CHAPTER 1 COMPLETE